Chaeun Park

chaeunnara@snu.ac.kr | Seoul, South Korea | chaephymath

Education

Seoul National University, Physics; Mathematical Sciences (Double Major) — Seoul, South Korea
- College of Natural Sciences — 2022 – 2027

Teaching

Topology Mentoring, Mentor — Seoul, South Korea
Private, in-person undergraduate mentoring based on Chapters 2–8 of Munkres' Topology. — Apr 2026 – Aug 2026

Algebra Mentoring, Mentor — Seoul, South Korea
Private, in-person mentoring for two students, covering selected topics from Chapters 1–14 of Dummit and Foote's Abstract Algebra. — Jan 2024 – Feb 2024

Academic Study Groups

Advanced Quantum Field Theory, Lead Organizer and Presenter — Seoul National University, Seoul, South Korea
Co-leading an advanced, year-long study of quantum field theory with a small group of core presenters. Topics are organized primarily around Ramond, with selected material from Schwartz, Srednicki, and other references. — Jan 2026 – present

Introductory Quantum Field Theory, Presenter — Seoul National University, Seoul, South Korea
Participated in a rotating seminar-style introduction to quantum field theory based on Ramond's Field Theory: A Modern Primer, with Professor Seok Kim invited as faculty mentor. — Jan 2025 – Feb 2025

Compact Lie Group Representation Theory, Presenter — Seoul National University, Seoul, South Korea
Participated in a rotating seminar based on Representations of Compact Lie Groups. — Sept 2024 – Dec 2024

Presentations

The Power of Symmetry: From Rotations to Particles, Forces, and Gravity — CERN, Geneva, Switzerland
Joint presentation with Drim Choi and Seokwon Lee at the CERN–CKC Summer Student Programme 2026. Presented Part III, "Massless Spin-2 and Einstein Gravity," on how gauge invariance and the self-consistent coupling of a massless spin-2 field lead from Fierz–Pauli theory to Einstein gravity. — July 2026

PIGE Analysis of $^{12}C(p,\gamma)^{13}N$ and $^{27}Al(p,\gamma)^{27}Al$, $Al(p,\alpha\gamma)^{24}Mg$ Reactions using Pelletron Accelerator — Seoul, South Korea
Poster presentation with Haeun Chung at the 2024 Undergraduate Research Presentation, based on experimental nuclear physics research conducted during Nishina School 2024. — Dec 2024

Awards

Excellence Award, 2024 Undergraduate Research Presentation — Seoul National University, Seoul, South Korea
Awarded by the Department of Physics and Astronomy for the poster "PIGE Analysis of $^{12}C(p,\gamma)^{13}N$ and $^{27}Al(p,\gamma)^{27}Al$, $Al(p,\alpha\gamma)^{24}Mg$ Reactions using Pelletron Accelerator," based on research conducted during Nishina School 2024. — Dec 2024

Conferences & Schools

The 6th International Undergraduate Mathematics Summer School — Seoul National University, Seoul, South Korea
Participant — 2026-07-27 to 2026-08-07

CERN-Korean Summer Student Program — CERN, Geneva, Switzerland
Participant — 2026-06-29 to 2026-07-11

SNU-APCTP Winter School on Fundamental Physics — Seoul National University, Seoul, South Korea
Participant — 2026-02-02 to 2026-02-06

The 12th Korea PDE Winter School — POSTECH, Pohang, South Korea
Participant — 2026-01-19 to 2026-01-23

Nishina School 2024 — RIKEN, Japan
Participant — 2024-07-25 to 2024-08-02

The 3rd International Undergraduate Mathematics Summer School — Seoul National University, Seoul, South Korea
Participant — 2023-08-07 to 2023-08-18

10-10 Summer School on Mathematics of Deep Learning and AI — Seoul National University, Seoul, South Korea
Participant — 2023-08-07 to 2023-08-08

Research Interests

Nonlinear PDEs and Fluids
Existence, regularity, and nonlinear dynamics of fluid equations, particularly the Euler equations, with attention to the effects of boundaries and singular geometry.

Hyperbolic Geometry and Geometric Group Theory
Discrete group actions on hyperbolic spaces, especially Fuchsian groups, and how group-theoretic and representation-theoretic properties are reflected in geometry, topology, and dynamics.

Quantum Field Theory and Gravitation
The relation between quantum field theory and spacetime geometry, from conserved quantities such as Komar integrals and ADM mass to holographic duality in AdS/CFT. I am also interested in how black-hole physics connects quantum chaos, random matrix theory, and SYK models.

Languages

Korean
Native

English
Academic working proficiency